**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

DAIRY FACILITY INFORMATION

**A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY:** Sousa Family Dairy

Physical address of dairy:

| Number and Street | City | County | Zip Code |
|---|---|---|---|
| 12801 W Bradbury RD | Turlock | Stanislaus | 95380 |

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1940

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

0058-0023-0015-0000 0058-0024-0008-0000

**B. OPERATORS**

Sousa, David

Operator name: Sousa, David

Telephone no.: Landline: ; Cellular: (209) 595-6452

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 12801 W Bradbury RD | Turlock | CA | 95380 |

**C. OWNERS**

Sousa, Armelim

Legal owner name: Sousa, Armelim

Telephone no.: Landline: (209) 632-5845; Cellular:

| Mailing Address Number and Street | City | State | Zip Code |
|---|---|---|---|
| 3030 S Central AVE | Turlock | CA | 95380 |

**This owner is responsible for paying permit fees.**

AVAILABLE NUTRIENTS

## A. HERD INFORMATION

| | Milk Cows | Dry Cows | Bred Heifers (15-24 mo.) | Heifers (7-14 mo. to breeding) | Calves (4-6 mo.) | Calves (0-3 mo.) |
|---|---|---|---|---|---|---|
| Number open confinement | 0 | 0 | 205 | 45 | 0 | 0 |
| Number under roof | 697 | 0 | 0 | 0 | 0 | 0 |
| Maximum number | 697 | 0 | 205 | 45 | 0 | 0 |
| Average number | 697 | 0 | 205 | 45 | 0 | 0 |
| Avg live weight (lbs) | 1,400 | 0 | 1,000 | 700 | | |

Predominant milk cow breed: Holstein

Average milk production: 90 *pounds per cow per day*

## B. MANURE GENERATED

Total manure excreted by the herd: 22,094.14 *tons per reporting period*

Total nitrogen from manure: 278,875.30 *lbs per reporting period*

After ammonia losses (30% loss applied): 195,212.71 *lbs per reporting period*

Total phosphorus from manure: 47,522.25 *lbs per reporting period*

Total potassium from manure: 142,640.51 *lbs per reporting period*

Total salt from manure: 328,182.45 *lbs per reporting period*

## C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 20,561,000 *gallons*

Total nitrogen generated: 89,583.47 *lbs*

Total phosphorus generated: 5,611.93 *lbs*

Total potassium generated: 112,708.17 *lbs*

Total salt generated: 774,001.25 *lbs*

| | | |
|---|---|---|
| | 20,561,000 | *gallons applied* |
| + | 0 | *gallons exported* |
| - | 0 | *gallons imported* |
| = | 20,561,000 | *gallons generated* |

## D. FRESH WATER SOURCES

| Source Description | Type |
|---|---|
| Turlock Irrigation District | Surface water |

## E. SUBSURFACE (TILE) DRAINAGE SOURCES

*No subsurface (tile) drainage sources entered.*

## F. NUTRIENT IMPORTS

*No dry manure nutrient imports entered.*

*No process wastewater nutrient imports entered.*

*No commercial or other nutrient imports entered.*

## G. NUTRIENT EXPORTS

*No solid nutrient exports entered.*

*No liquid nutrient exports entered.*

APPLICATION AREA

## A. LIST OF LAND APPLICATION AREAS

| Field name | Controlled acres | Cropable acres | Total harvests | Type of waste applied | Parcel number |
|---|---|---|---|---|---|
| Dairy Field | 100 | 100 | 2 | process wastewater | X058-X023-X015-XXXX<br>X058-X024-X005-XXXX<br>X058-X024-X008-XXXX |
| Field 140 | 135 | 135 | 1 | manure | X057-X001-X013-XXXX |
| Field 3 & 4 | 93 | 93 | 2 | process wastewater | X058-X023-X013-XXXX<br>X058-X023-X014-XXXX |
| Totals for areas that were used for application | 328 | 328 | 5 | | |
| Totals for areas that were not used for application | | | | | |
| Land application area totals | 328 | 328 | 5 | | |

## B. CROPS AND HARVESTS

Dairy Field

Field name: Dairy Field

11/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 100 Plant date: 11/11/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/17/2023 | 1,587.00 *ton* | Dry-weight | | 63.8 | 15,400.00 | 2,970.00 | 27,900.00 | | 11.10 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 1,600.00 |
| Total actual harvest content | 15.87 | 176.94 | 34.12 | 320.57 | 1,275.38 |

## Dairy Field

### 06/12/2023: Corn, silage

Crop: Corn, silage  Acres planted: 100  Plant date: 06/12/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/28/2023 | 2,887.00 *ton* | Dry-weight | | 67.9 | 10,600.00 | 2,240.00 | 14,900.00 | | 5.32 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 1,500.00 |
| Total actual harvest content | 28.87 | 196.47 | 41.52 | 276.16 | 986.04 |

## Field 140

Field name: Field 140

### 10/25/2020: Alfalfa, hay

Crop: Alfalfa, hay  Acres planted: 135  Plant date: 10/25/2020

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 07/08/2023 | 1,080.00 *ton* | Dry-weight | | 8.3 | 33,900.00 | 2,000.00 | 15,300.00 | | 5.00 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 8.00 | 480.00 | 43.20 | 336.00 | 800.00 |
| Total actual harvest content | 8.00 | 497.38 | 29.34 | 224.48 | 733.60 |

## Field 3 & 4

Field name: Field 3 & 4

## Field 3 & 4

### 11/06/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 93 Plant date: 11/06/2022

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 05/15/2023 | 1,395.00 *ton* | Dry-weight | | 74.2 | 18,300.00 | 2,820.00 | 28,000.00 | | 10.60 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 16.00 | 160.00 | 25.60 | 132.80 | 1,600.00 |
| Total actual harvest content | 15.00 | 141.64 | 21.83 | 216.72 | 820.44 |

### 05/16/2023: Corn, silage

Crop: Corn, silage Acres planted: 93 Plant date: 05/16/2023

| Harvest date | Yield | Reporting basis | Density (lbs/cu ft) | Moisture (%) | N (mg/kg) | P (mg/kg) | K (mg/kg) | Salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|
| 09/12/2023 | 2,763.00 *ton* | Dry-weight | | 60.4 | 12,800.00 | 1,360.00 | 11,800.00 | | 5.68 |

| | Yield (tons/acre) | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Salt (lbs/acre) |
|---|---|---|---|---|---|
| Anticipated harvest content | 30.00 | 240.00 | 45.00 | 198.00 | 1,500.00 |
| Total actual harvest content | 29.71 | 301.18 | 32.00 | 277.65 | 1,336.51 |

NUTRIENT BUDGET

## A. LAND APPLICATIONS

### Dairy Field - 11/11/2022: Oats, silage-soft dough

Field name: Dairy Field

Crop: Oats, silage-soft dough — Plant date: 11/11/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/15/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 95.98 | 4.80 | 105.93 | 662.94 | 1,727,000.00 *gal* |
| Turlock Irrigation District | Surface water | 2.23 | 0.00 | 0.00 | 40.49 | 16,729,000.00 *gal* |
| Application event totals | | 98.22 | 4.80 | 105.93 | 703.43 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 04/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 110.43 | 5.52 | 121.87 | 762.75 | 1,987,000.00 *gal* |
| Turlock Irrigation District | Surface water | 2.04 | 0.00 | 0.00 | 37.02 | 15,297,000.00 *gal* |
| Application event totals | | 112.48 | 5.52 | 121.87 | 799.77 | |

### Dairy Field - 06/12/2023: Corn, silage

Field name: Dairy Field

Crop: Corn, silage — Plant date: 06/12/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 33.79 | 2.47 | 49.29 | 346.04 | 978,000.00 *gal* |
| Turlock Irrigation District | Surface water | 1.79 | 0.00 | 0.00 | 32.49 | 13,427,000.00 *gal* |
| Application event totals | | 35.58 | 2.47 | 49.29 | 378.54 | |

## Dairy Field - 06/12/2023: Corn, silage

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 1.87 | 0.00 | 0.00 | 33.87 | 13,995,000.00 *gal* |
| Application event totals | | 1.87 | 0.00 | 0.00 | 33.87 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 66.89 | 4.90 | 97.58 | 685.01 | 1,936,000.00 *gal* |
| Turlock Irrigation District | Surface water | 1.69 | 0.00 | 0.00 | 30.55 | 12,624,000.00 *gal* |
| Application event totals | | 68.57 | 4.90 | 97.58 | 715.56 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/17/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 72.46 | 4.97 | 83.45 | 657.10 | 1,497,000.00 *gal* |
| Turlock Irrigation District | Surface water | 1.65 | 0.00 | 0.00 | 29.90 | 12,356,951.00 *gal* |
| Application event totals | | 74.11 | 4.97 | 83.45 | 687.01 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/02/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 75.70 | 5.19 | 87.18 | 686.51 | 1,564,000.00 *gal* |
| Turlock Irrigation District | Surface water | 1.59 | 0.00 | 0.00 | 28.85 | 11,920,000.00 *gal* |
| Application event totals | | 77.29 | 5.19 | 87.18 | 715.36 | |

## Field 140 - 10/25/2020: Alfalfa, hay

Field name: Field 140

Crop: Alfalfa, hay

Plant date: 10/25/2020

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|

Field 140 - 10/25/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.38 | 0.00 | 0.00 | 43.05 | 24,016,000.00 *gal* |
| Application event totals | | 2.38 | 0.00 | 0.00 | 43.05 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/03/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.49 | 0.00 | 0.00 | 45.13 | 25,173,000.00 *gal* |
| Application event totals | | 2.49 | 0.00 | 0.00 | 45.13 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/06/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.40 | 0.00 | 0.00 | 43.48 | 24,256,000.00 *gal* |
| Application event totals | | 2.40 | 0.00 | 0.00 | 43.48 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.17 | 0.00 | 0.00 | 39.31 | 21,927,000.00 *gal* |
| Application event totals | | 2.17 | 0.00 | 0.00 | 39.31 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/18/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.30 | 0.00 | 0.00 | 41.73 | 23,278,000.00 *gal* |
| Application event totals | | 2.30 | 0.00 | 0.00 | 41.73 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 09/16/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.34 | 0.00 | 0.00 | 42.33 | 23,614,000.00 *gal* |
| Application event totals | | 2.34 | 0.00 | 0.00 | 42.33 | |

## Field 140 - 10/25/2020: Alfalfa, hay

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/22/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.18 | 0.00 | 0.00 | 39.43 | 21,993,000.00 *gal* |
| Application event totals | | 2.18 | 0.00 | 0.00 | 39.43 | |

| | | | | |
|---|---|---|---|---|
| 10/30/2023 | Broadcast/incorporate | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Manure | Corral solids | 196.39 | 148.53 | 402.31 | 14,967.45 | 1,790.00 *ton* |
| Application event totals | | 196.39 | 148.53 | 402.31 | 14,967.45 | |

## Field 3 & 4 - 11/06/2022: Oats, silage-soft dough

Field name: Field 3 & 4

Crop: Oats, silage-soft dough

Plant date: 11/06/2022

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 10/16/2022 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 91.25 | 4.56 | 100.71 | 630.29 | 1,527,000.00 *gal* |
| Turlock Irrigation District | Surface water | 2.14 | 0.00 | 0.00 | 38.75 | 14,890,000.00 *gal* |
| Application event totals | | 93.39 | 4.56 | 100.71 | 669.04 | |

| | | | | |
|---|---|---|---|---|
| 04/08/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 93.41 | 4.67 | 103.08 | 645.15 | 1,563,000.00 *gal* |
| Turlock Irrigation District | Surface water | 2.20 | 0.00 | 0.00 | 39.79 | 15,289,000.00 *gal* |
| Application event totals | | 95.60 | 4.67 | 103.08 | 684.93 | |

## Field 3 & 4 - 05/16/2023: Corn, silage

## Field 3 & 4 - 05/16/2023: Corn, silage

Field name: Field 3 & 4

Crop: Corn, silage

Plant date: 05/16/2023

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 05/07/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 54.09 | 3.96 | 78.91 | 553.95 | 1,456,000.00 *gal* |
| Turlock Irrigation District | Surface water | 2.17 | 0.00 | 0.00 | 39.30 | 15,102,000.00 *gal* |
| Application event totals | | 56.26 | 3.96 | 78.91 | 593.25 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/01/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 71.59 | 5.24 | 104.44 | 733.15 | 1,927,000.00 *gal* |
| Turlock Irrigation District | Surface water | 2.00 | 0.00 | 0.00 | 36.24 | 13,927,000.00 *gal* |
| Application event totals | | 73.59 | 5.24 | 104.44 | 769.39 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 06/30/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 2.09 | 0.00 | 0.00 | 37.79 | 14,523,000.00 *gal* |
| Application event totals | | 2.09 | 0.00 | 0.00 | 37.79 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 07/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 84.10 | 6.16 | 122.70 | 861.36 | 2,264,000.00 *gal* |
| Turlock Irrigation District | Surface water | 1.99 | 0.00 | 0.00 | 36.14 | 13,887,000.00 *gal* |
| Application event totals | | 86.10 | 6.16 | 122.70 | 897.50 | |

| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/14/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Lagoon | Process wastewater | 79.31 | 5.80 | 115.71 | 812.28 | 2,135,000.00 *gal* |
| Turlock Irrigation District | Surface water | 2.01 | 0.00 | 0.00 | 36.45 | 14,008,000.00 *gal* |
| Application event totals | | 81.32 | 5.80 | 115.71 | 848.73 | |

Field 3 & 4 - 05/16/2023: Corn, silage

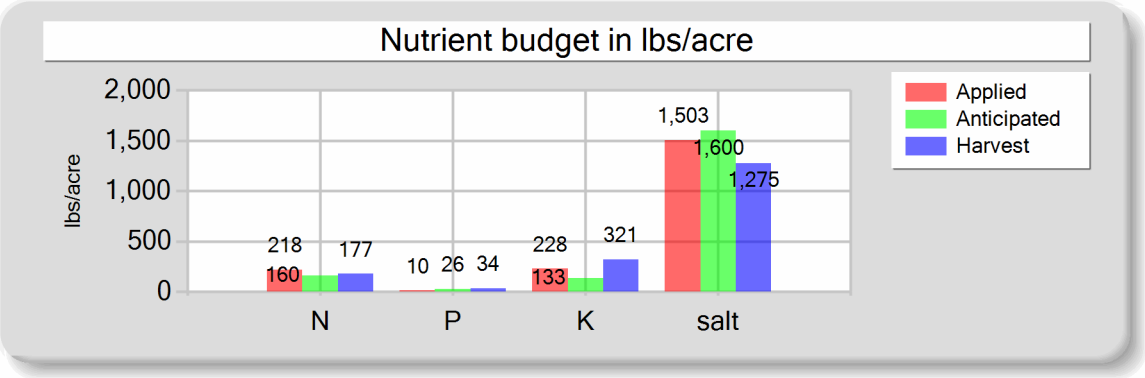
| Application date | Application method | Precipitation 24 hours prior | Precipitation during application | Precipitation 24 hours following |
|---|---|---|---|---|
| 08/28/2023 | Surface (irrigation) | No precipitation | No precipitation | No precipitation |

| Source description | Material type | N (lbs/acre) | P (lbs/acre) | K (lbs/acre) | Salt (lbs/acre) | Amount |
|---|---|---|---|---|---|---|
| Turlock Irrigation District | Surface water | 1.95 | 0.00 | 0.00 | 35.30 | 13,567,000.00 *gal* |
| Application event totals | | 1.95 | 0.00 | 0.00 | 35.30 | |

## B. NUTRIENT BUDGET

Dairy Field - 11/11/2022: Oats, silage-soft dough

Field name: Dairy Field  Crop: Oats, silage-soft dough  Plant date: 11/11/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 206.42 | 10.32 | 227.80 | 1,425.69 |
| Fresh water | 4.28 | 0.00 | 0.00 | 77.50 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 217.69 | 10.32 | 227.80 | 1,503.20 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 1,600.00 |
| Actual crop nutrient removal | 176.94 | 34.12 | 320.57 | 1,275.38 |
| Nutrient balance | 40.75 | -23.80 | -92.77 | 227.82 |
| Applied to removed ratio | 1.23 | 0.30 | 0.71 | 1.18 |

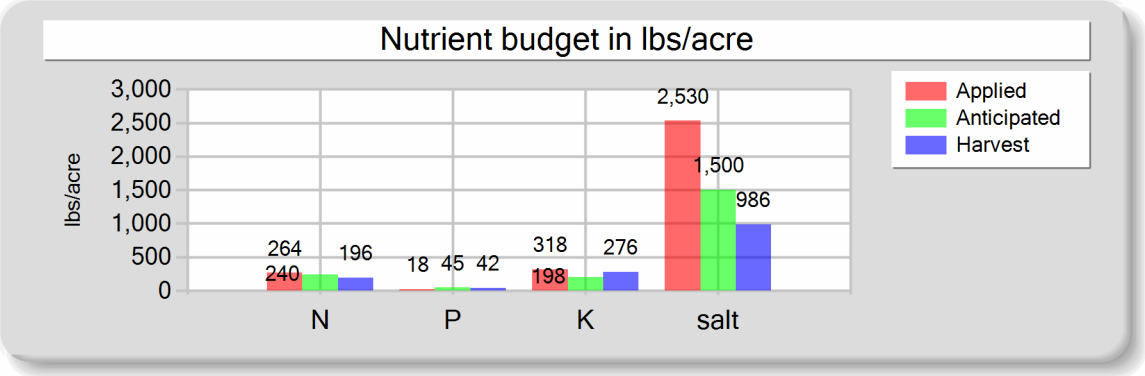
| Fresh water applied | |
|---|---|
| 32,026,000.00 | *gallons* |
| 1,179.41 | *acre-inches* |
| 11.79 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,714,000.00 | *gallons* |
| 136.77 | *acre-inches* |
| 1.37 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Dairy Field - 06/12/2023: Corn, silage

Field name: Dairy Field  Crop: Corn, silage  Plant date: 06/12/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 248.83 | 17.53 | 317.51 | 2,374.67 |
| Fresh water | 8.59 | 0.00 | 0.00 | 155.66 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 264.42 | 17.53 | 317.51 | 2,530.34 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 1,500.00 |
| Actual crop nutrient removal | 196.47 | 41.52 | 276.16 | 986.04 |
| Nutrient balance | 67.95 | -23.98 | 41.35 | 1,544.30 |
| Applied to removed ratio | 1.35 | 0.42 | 1.15 | 2.57 |

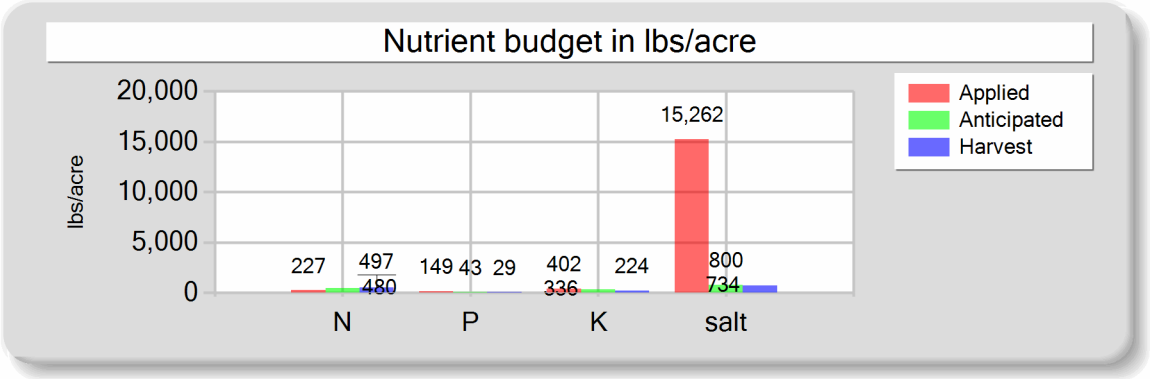
| Fresh water applied | |
|---|---|
| 64,322,951.00 | *gallons* |
| 2,368.80 | *acre-inches* |
| 23.69 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 5,975,000.00 | *gallons* |
| 220.04 | *acre-inches* |
| 2.20 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 140 - 10/25/2020: Alfalfa, hay

Field name: Field 140 Crop: Alfalfa, hay Plant date: 10/25/2020




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 196.39 | 148.53 | 402.31 | 14,967.45 |
| Process wastewater | 0.00 | 0.00 | 0.00 | 0.00 |
| Fresh water | 16.25 | 0.00 | 0.00 | 294.45 |
| Atmospheric deposition | 14.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 226.63 | 148.53 | 402.31 | 15,261.90 |
| Anticipated crop nutrient removal | 480.00 | 43.20 | 336.00 | 800.00 |
| Actual crop nutrient removal | 497.38 | 29.34 | 224.48 | 733.60 |
| Nutrient balance | -270.75 | 119.19 | 177.83 | 14,528.30 |
| Applied to removed ratio | 0.46 | 5.06 | 1.79 | 20.80 |

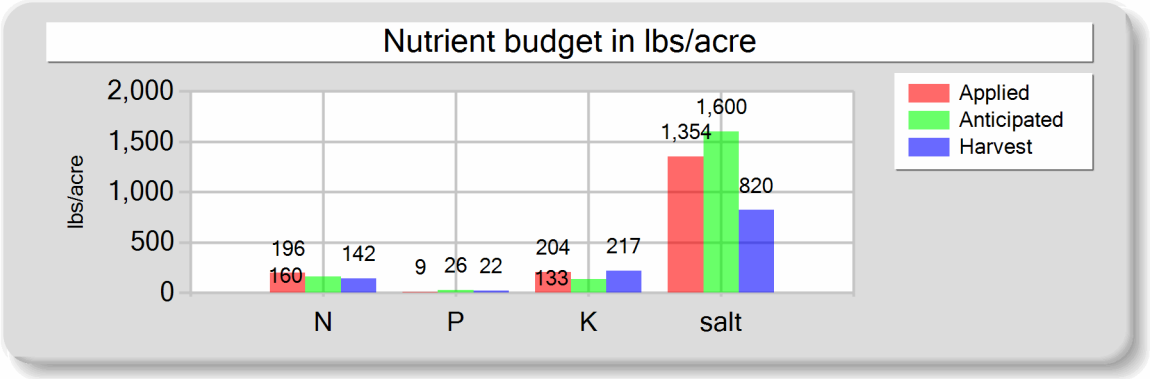
| Fresh water applied | |
|---|---|
| 164,257,000.00 | *gallons* |
| 6,049.03 | *acre-inches* |
| 44.81 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 0.00 | *gallons* |
| 0.00 | *acre-inches* |
| 0.00 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 & 4 - 11/06/2022: Oats, silage-soft dough

Field name: Field 3 & 4 Crop: Oats, silage-soft dough Plant date: 11/06/2022




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 184.66 | 9.23 | 203.79 | 1,275.44 |
| Fresh water | 4.33 | 0.00 | 0.00 | 78.53 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 195.99 | 9.23 | 203.79 | 1,353.97 |
| Anticipated crop nutrient removal | 160.00 | 25.60 | 132.80 | 1,600.00 |
| Actual crop nutrient removal | 141.64 | 21.83 | 216.72 | 820.44 |
| Nutrient balance | 54.35 | -12.59 | -12.93 | 533.53 |
| Applied to removed ratio | 1.38 | 0.42 | 0.94 | 1.65 |

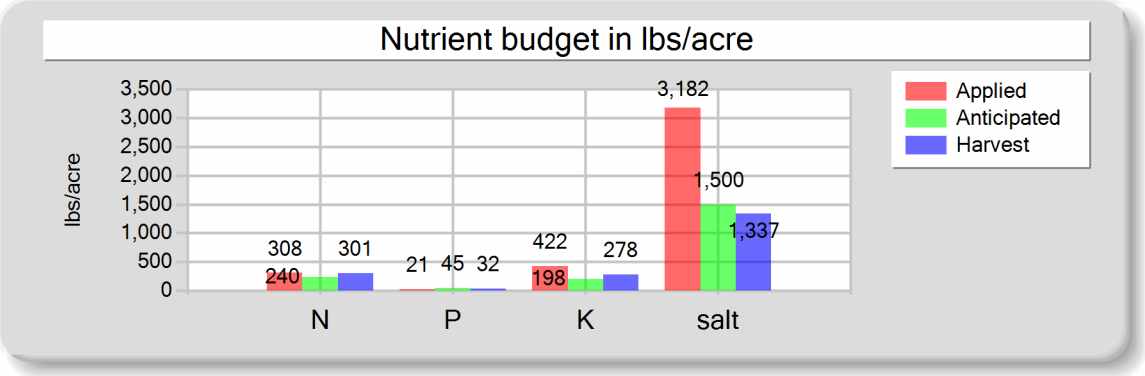
| Fresh water applied | |
|---|---|
| 30,179,000.00 | *gallons* |
| 1,111.39 | *acre-inches* |
| 11.95 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 3,090,000.00 | *gallons* |
| 113.79 | *acre-inches* |
| 1.22 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

## Field 3 & 4 - 05/16/2023: Corn, silage

Field name: Field 3 & 4 Crop: Corn, silage Plant date: 05/16/2023




| | Total N (lbs/acre) | Total P (lbs/acre) | Total K (lbs/acre) | Total salt (lbs/acre) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 0.00 | 0.00 | 0.00 | 0.00 |
| Process wastewater | 289.09 | 21.16 | 421.77 | 2,960.74 |
| Fresh water | 12.21 | 0.00 | 0.00 | 221.22 |
| Atmospheric deposition | 7.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 308.30 | 21.16 | 421.77 | 3,181.97 |
| Anticipated crop nutrient removal | 240.00 | 45.00 | 198.00 | 1,500.00 |
| Actual crop nutrient removal | 301.18 | 32.00 | 277.65 | 1,336.51 |
| Nutrient balance | 7.11 | -10.84 | 144.11 | 1,845.46 |
| Applied to removed ratio | 1.02 | 0.66 | 1.52 | 2.38 |

| Fresh water applied | |
|---|---|
| 85,014,000.00 | *gallons* |
| 3,130.78 | *acre-inches* |
| 33.66 | *inches/acre* |

| Process wastewater applied | |
|---|---|
| 7,782,000.00 | *gallons* |
| 286.58 | *acre-inches* |
| 3.08 | *inches/acre* |

| Total harvests for the crop | |
|---|---|
| 1 | *harvests* |

NUTRIENT ANALYSES

## A. MANURE ANALYSES

### T4002005D

Sample and source description: T4002005D

Sample date: 10/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 28.1 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 10,300.00 | 7,790.00 | 21,100.00 | 7,400.00 | 3,360.00 | 3,020.00 | 2,360.00 | 595.00 | | 78.50 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | | 0.10 |

### T4002005C

Sample and source description: T4002005C

Sample date: 11/15/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 23.5 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Calcium (mg/kg) | Magnesium (mg/kg) | Sodium (mg/kg) | Sulfur (mg/kg) | Chloride (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 8,090.00 | 7,950.00 | 20,300.00 | 8,430.00 | 3,440.00 | 2,910.00 | 2,310.00 | 648.00 | | 70.40 |
| **DL** | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | 100.00 | | 0.10 |

## B. PROCESS WASTEWATER ANALYSES

### T3184016B

Sample and source description: T3184016B

Sample date: 03/08/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 666.00 | 462.00 | 0.00 | 0.00 | 33.30 | 735.00 | | | | | | | | 8,970.00 | 4,600 |
| **DL** | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | | | | | | | | 20.00 | 10 |

T3184016A

Sample and source description: T3184016A

Sample date: 06/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 414.00 | 283.00 | 0.00 | 0.00 | 30.30 | 604.00 | | | | | | | | 6,810.00 | 4,240 |
| **DL** | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | | | | | | | | 20.00 | 10 |

T4002005B

Sample and source description: T4002005B

Sample date: 08/18/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 580.00 | 300.00 | 0.00 | 0.00 | 39.80 | 668.00 | | | | | | | | 79,700.00 | 5,260 |
| **DL** | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | | | | | | | | 20.00 | 10 |

T4002005A

Sample and source description: T4002005A

Sample date: 12/15/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

| | Kjeldahl-N (mg/L) | NH4-N (mg/L) | NH3-N (mg/L) | Nitrate-N (mg/L) | Total P (mg/L) | Total K (mg/L) | Calcium (mg/L) | Magnes. (mg/L) | Sodium (mg/L) | Bicarb. (mg/L) | Carb. (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 739.00 | 336.00 | 0.00 | 0.00 | 66.90 | 712.00 | | | | | | | | 9,380.00 | 6,140 |
| **DL** | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | 0.10 | | | | | | | | 20.00 | 10 |

## C. FRESH WATER ANALYSES

Turlock Irrigation District

Turlock Irrigation District

TID info

Sample description: TID info

Sample date: 06/20/2023 Source of analysis: Lab analysis

| | Total N (mg/L) | NH4-N (mg/L) | Nitrate-N (mg/L) | Calcium (mg/L) | Magnesium (mg/L) | Sodium (mg/L) | Bicarbonate (mg/L) | Carbonate (mg/L) | Sulfate (mg/L) | Chloride (mg/L) | EC (µmhos/cm) | TDS (mg/L) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Value** | 1.60 | 0.00 | 1.60 | | | | | | | | 49.00 | 29 |
| **DL** | 0.10 | 0.10 | 0.10 | | | | | | | | 20.00 | 10 |

## D. SOIL ANALYSES

*No soil analyses entered.*

## E. PLANT TISSUE ANALYSES

Dairy Field - 11/11/2022: Oats, silage-soft dough

T3144038C

Sample and source description: T3144038C

Sample date: 05/17/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 63.8 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 15,400.00 | 2,970.00 | 27,900.00 | | 11.10 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.10 |

Dairy Field - 06/12/2023: Corn, silage

## Dairy Field - 06/12/2023: Corn, silage

### T3278039C

Sample and source description: T3278039C

Sample date: 09/28/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 67.9 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 10,600.00 | 2,240.00 | 14,900.00 | | 5.32 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.10 |

## Field 3 & 4 - 11/06/2022: Oats, silage-soft dough

### T3144038A

Sample and source description: T3144038A

Sample date: 05/15/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.2 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 18,300.00 | 2,820.00 | 28,000.00 | | 10.60 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.10 |

## Field 3 & 4 - 05/16/2023: Corn, silage

### T3261044A

Sample and source description: T3261044A

Sample date: 09/12/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.4 %

| | Total N (mg/kg) | Total P (mg/kg) | Total K (mg/kg) | Total salt (mg/kg) | TFS (%) |
|---|---|---|---|---|---|
| **Value** | 12,800.00 | 1,360.00 | 11,800.00 | | 5.68 |
| **DL** | 100.00 | 100.00 | 100.00 | | 0.10 |

## F. SUBSURFACE (TILE) DRAINAGE ANALYSES

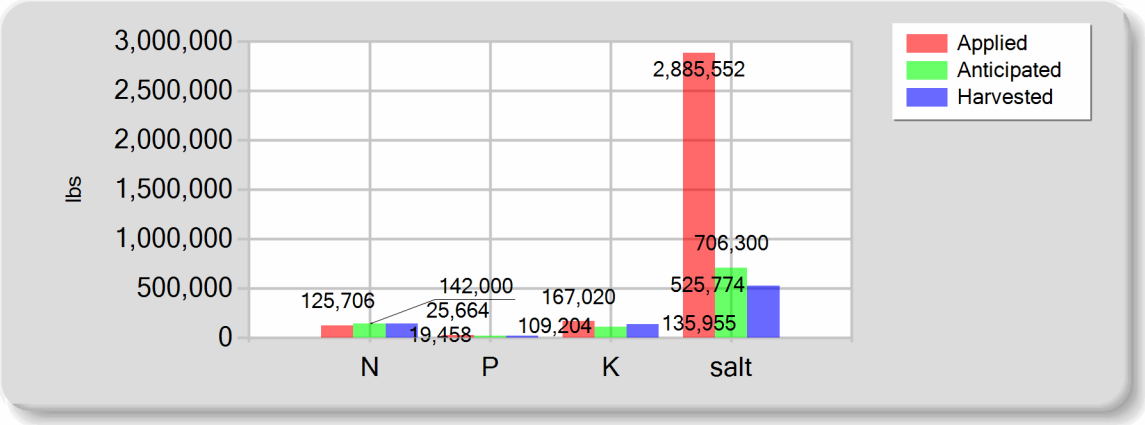
*No subsurface (tile) drainage analyses entered.*

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

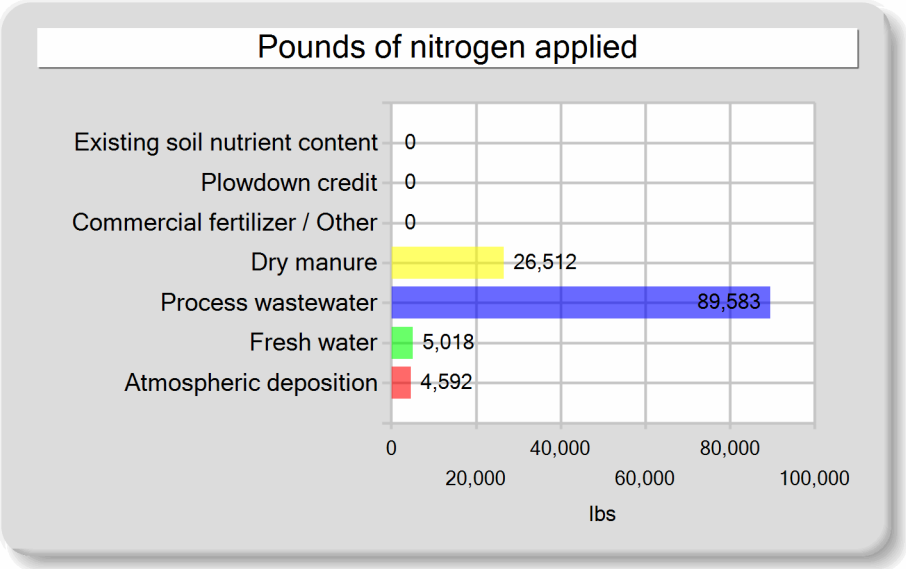
## A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

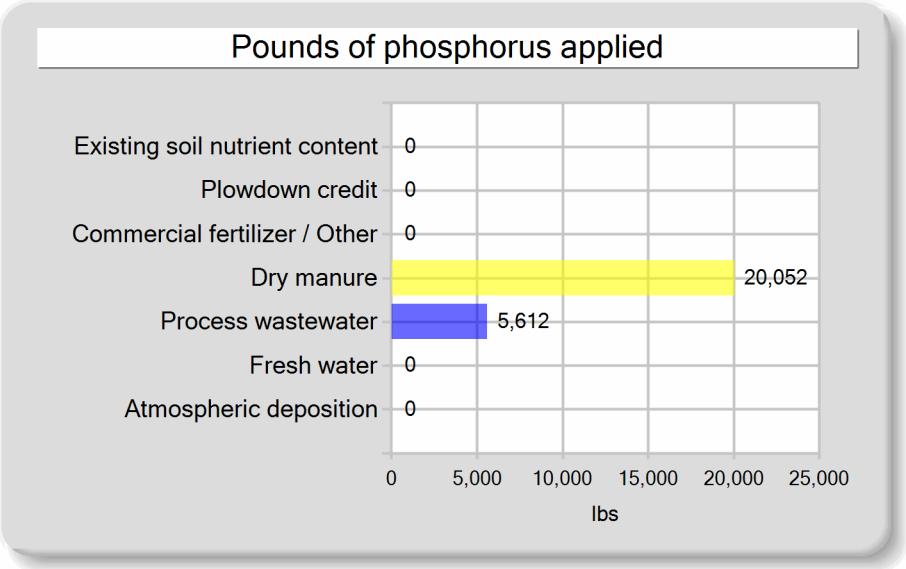
| | Total N (lbs) | Total P (lbs) | Total K (lbs) | Total salt (lbs) |
|---|---|---|---|---|
| Existing soil nutrient content | 0.00 | 0.00 | 0.00 | 0.00 |
| Plowdown credit | 0.00 | 0.00 | 0.00 | 0.00 |
| Commercial fertilizer / Other | 0.00 | 0.00 | 0.00 | 0.00 |
| Dry manure | 26,512.41 | 20,051.62 | 54,311.82 | 2,020,605.70 |
| Process wastewater | 89,583.47 | 5,611.93 | 112,708.17 | 774,001.25 |
| Fresh water | 5,017.67 | 0.00 | 0.00 | 90,945.23 |
| Atmospheric deposition | 4,592.00 | 0.00 | 0.00 | 0.00 |
| Total nutrients applied | 125,705.55 | 25,663.54 | 167,019.99 | 2,885,552.18 |
| Anticipated crop nutrient removal | 142,000.00 | 19,457.80 | 109,204.40 | 706,300.00 |
| Actual crop nutrient removal | 145,670.33 | 16,531.65 | 135,955.10 | 525,773.55 |
| Nutrient balance | -19,964.78 | 9,131.90 | 31,064.89 | 2,359,778.63 |
| Applied to removed ratio | 0.86 | 1.55 | 1.23 | 5.49 |

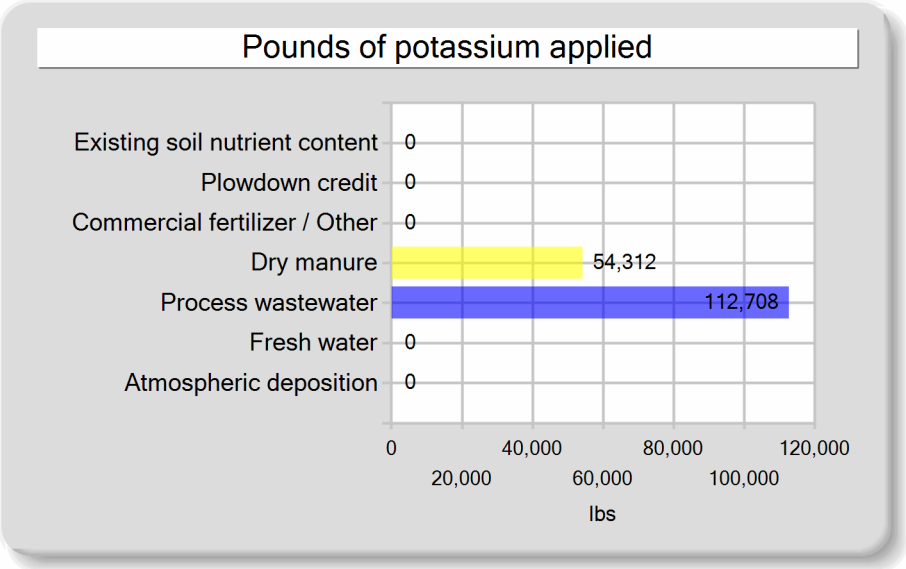
## B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

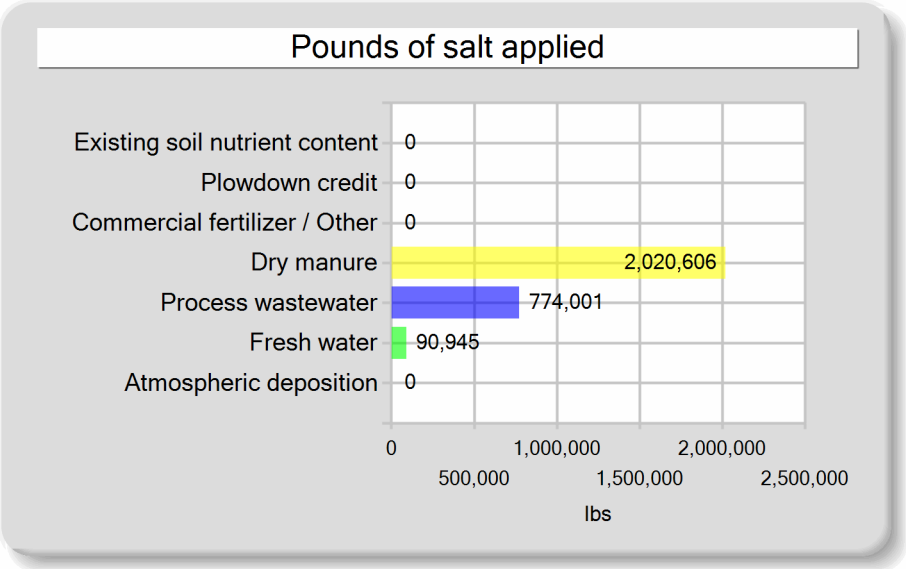


## C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE



## EXCEPTION REPORTING

### A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

*No manure or process wastewater discharges occurred during the reporting period.*

### B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

*No stormwater discharges occurred during the reporting period.*

### C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

*No land application area to surface water discharges occurred during the reporting period.*

## NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

### A. NUTRIENT MANAGEMENT PLAN STATEMENTS

| | |
|---|---|
| Was the facility's NMP updated in the reporting period? | No |
| Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |
| Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order? | Yes |

### B. EXPORT AGREEMENT STATEMENT

| | |
|---|---|
| Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period? | No |

ADDITIONAL NOTES

**A. NOTES**

*No notes entered for this annual report.*

**Annual Report - General Order No. R5-2007-0035**
*Reporting period 01/01/2023 to 12/31/2023.*

CERTIFICATION

## A. OWNER AND/OR OPERATOR CERTIFICATION

*I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.*

| | |
|---|---|
| Armelim de Sousa | |
| SIGNATURE OF OWNER OF FACILITY | SIGNATURE OF OPERATOR OF FACILITY |
| Armelim Sousa | David Sousa |
| PRINT OR TYPE NAME | PRINT OR TYPE NAME |
| 6-28-24 | 6-28-24 |
| DATE | DATE |

ATTACHMENTS

## A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

| Annual Dairy Facility Assessment |
| --- |
| Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A ) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed . |

| Manure/Process Wastewater Tracking Manifests |
| --- |
| Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler. |

| Corrective Actions Documents |
| --- |
| Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction. |

| Groundwater Monitoring |
| --- |
| Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13. |

| Storm Water Monitoring |
| --- |
| Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14. |

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

# Compliance Analysis Report

**Customer**
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

**Grower**

**Consultant:**
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 11/13/23 1:08 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 11/16/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| Well 5b | T3317010A | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 0.515 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Nitrate Nitrogen (NO3-N) | 99.5 | 98.9 (T3317005A) | 98.7 (T3317005A) | 0.223 |

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.



ELAP Certificate No.: 2714

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

# Compliance Analysis Report

**Customer**
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

**Grower**

**Consultant:**
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 11/13/23 1:08 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 11/16/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| Well 5c | T3317010B | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 0.154 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Nitrate Nitrogen (NO3-N) | 99.5 | 98.9 (T3317005A) | 98.7 (T3317005A) | 0.223 |

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# Compliance Analysis Report

**Customer**
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

**Grower**

**Consultant:**
Paul Sousa

PURCHASE ORDER:
RECEIVED DATE: 11/13/23 1:08 PM
SUBMITTED BY: Paul Sousa
ANALYZED DATE: 11/16/23

Nitrate Nitrogen (NO3-N)
SOURCE: Well Water

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| Well 7 | T3317010C | Nitrate Nitrogen (NO3-N) | EPA 353.2 | 2.42 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Nitrate Nitrogen (NO3-N) | 99.5 | 98.9 (T3317005A) | 98.7 (T3317005A) | 0.223 |

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# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380

PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: A & M de Sousa Dairy
Mailing address:
(not facility address)
Phone#:

Authorized Copy Released To: Paul Sousa

Sampled By: Paul Sousa
Project ID: Sousa Family Dairy

## ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
- H1 EC*, NO3-N, NH4-N*
- H6 EC, TN, TDS
- H7 Combination of H1 & H6
- H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
- H3 EC*, NH4-N, TKN, TP, K, TDS
- H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- H8 Combination of H3 &H5
- H9 Combination of H4 &H5

**TILE DRAIN**
- H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
- E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
- M1 % Moisture
- M3 TN TP, K, % Moisture
- M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- M7 Combination of M3 & M4

**PLANT TISSUE**
- F18 % Moisture, Ash, TN, P, K, DM and as received
- F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- F19+ % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
- K3 Soluble Phosphorus
- K4 1ft NO3, Organic Matter
- K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
- K7 Combination of K3 & K4
- K8 Combination of K3 & K5

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | Analysis: H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 11/12/23 | 3:10 | Well 5b | | 2.0°C | | $NO_3$-N | | | | |
| 2 | 11/12/23 | 3:15 | Well 5c | | 1.0°C | | $NO_3$-N | | | | |
| 3 | 11/12/23 | 3:25 | Well 7 | | 1.6°C | | $NO_3$-N | | | | |
| 4 | | | | | | | | | | | |
| 5 | | | | | | | | | | | |
| 6 | | | | | | | | | | | |
| 7 | | | | | | | | | | | |
| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

**CHAIN OF CUSTODY RECORDING**

| | Signature | Company | Received Date & Time | Relinquished Date & Time | (LABORATORY USE ONLY) Logged in By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | Paul Sousa | | | | | |
| 2nd | | DENELE ANALYTICAL, INC. | 11-13-23 1:08 | | | T3317010 |

Required to be submitted to any agency? (YES / NO)

**Denele Analytical, Inc.**

865 South Kilroy Avenue Turlock, CA 95380
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40 N. East Street, Woodland, CA 95776
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www.denelelabs.com

# Compliance Analysis Report

**Customer**
A & M De Sousa Dairy
3030 S. Central Ave.
Turlock, CA 95380

**Grower**

**Consultant:**
Paul Sousa

PURCHASE ORDER: Sousa Family Dairy
RECEIVED DATE: 10/16/23 3:27 PM
SUBMITTED BY: Paul
ANALYZED DATE: 10/17/23

Nitrate Nitrogen (NO3-N)
SOURCE: Irrigation Water

| Sample ID | Lab ID | Analyte | Method Ref | Result | Units |
|---|---|---|---|---|---|
| SFD Ag Well | T3290008A | Nitrate Nitrogen (NO3-N) | EPA 353.2 | < 0.140 | mg/L |

| Laboratory Quality Control | LCS % | MS % | MSD % | RPD % |
|---|---|---|---|---|
| Nitrate Nitrogen (NO3-N) | 98.8 | 101 (W3289001A) | 103 (W3289001A) | 1.35 |

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ELAP Certificate No.: 2714

# DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

## DAIRY COMPLIANCE

Client: A&M De Sousa
Mailing address: 3030 S Central Ave
(not facility address) Turlock 95380
Phone#:

Authorized Copy Released To: Paul Sousa

Sampled By: Paul
Project ID: Sousa Family Dairy

## ANALYSIS TO BE COMPLETED:

**IRRIGATION/GROUND WATER**
- **H1** EC*, NO3-N, NH4-N*
- **H6** EC, TN, TDS
- **H7** Combination of H1 & H6
- **H14** EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

**PROCESS WASTE WATER**
- **H3** EC*, NH4-N, TKN, TP, K, TDS
- **H4** EC*, NO3-N, NH4-N, TKN, TP, K, TDS
- **H5** Ca, Mg, Na, Bicarb, Carb, SO4, Cl
- **H8** Combination of H3 &H5
- **H9** Combination of H4 &H5

**TILE DRAIN**
- **H2** EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*

___ (Check here if performed)

**DISCHARGE WATER**
- **E1** EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
- **E2** EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
- **E3** EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
- **E4** EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

**MANURE**
- **M1** % Moisture
- **M3** TN TP, K, % Moisture
- **M4** Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
- **M7** Combination of M3 & M4

**PLANT TISSUE**
- **F18** % Moisture, Ash, TN, P, K, DM and as received
- **F18+** % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
- **F19+** % Moisture, TN (Package includes complete plant tissue analysis)

**SOIL**
- **K3** Soluble Phosphorus
- **K4** 1ft NO3, Organic Matter
- **K5** Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
- **K6** 0-1ft EC, NO3, Soluble P, K, Organic Matter
- **K7** Combination of K3 & K4
- **K8** Combination of K3 & K5

| H1 | H3 | M3 | F18 | OTHER |
|---|---|---|---|---|
| | | | | |

| | Date Collected | Time Collected | Sample ID | Description | Temp | Well Purge | Analysis | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 10/15/23 | 4:55 | SFD Ag Well | | 4.2'c | | $NO_3$-N | | | | |
| 2 | | | | | | | | | | | |
| 3 | | | | | | | | | | | |
| 4 | | | | | | | | | | | |
| 5 | | | | | | | | | | | |
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| 8 | | | | | | | | | | | |
| 9 | | | | | | | | | | | |
| 10 | | | | | | | | | | | |

## CHAIN OF CUSTODY RECORDING

| | Signature | Company | Received Date & Time | Relinquished Date & Time | (LABORATORY USE ONLY) Logged in By: | LABORATORY NUMBER |
|---|---|---|---|---|---|---|
| 1st | Paul Sousa | | | | | |
| 2nd | | DENELE ANALYTICAL, INC. | 10/16/23 @ 15:27 | | | T3290008 |

Required to be submitted to any agency? (YES / NO)